Vocabulary: Causal Inference and ML: From Graphs to Effects

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Contents

1. Causality Foundations

- **Pearl's Ladder of Causation**: A three-level hierarchy distinguishing the depth of causal reasoning:
  - **Level 1 — Association**: Observing patterns. *"If I see X, what is Y?"* Standard ML operates here.
  - **Level 2 — Intervention**: Acting on the world. *"What happens if I do X?"* Requires the do-operator.
  - **Level 3 — Counterfactual**: Imagining alternatives. *"What if I had done X instead?"* Requires potential outcomes.

  Insight: Standard machine learning operates at Level 1 only. Levels 2 and 3 require explicit causal assumptions.

- **Potential Outcomes Framework (Rubin)**: For each unit i, define $Y_i(1)$ and $Y_i(0)$ as the outcomes under treatment and control respectively.

$$\tau_i = Y_i(1) - Y_i(0)$$

  Insight: Individual treatment effects τ_i are never directly observable — we only ever observe one realised outcome per unit.

- **Fundamental Problem of Causal Inference**: We can only observe one potential outcome per unit:

$$Y_i^{\text{obs}} = W_i \cdot Y_i(1) + (1 - W_i) \cdot Y_i(0)$$

  The counterfactual outcome remains forever unobserved.

- **Average Treatment Effect (ATE)**: The population-average causal effect of treatment:

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$$

- **Conditional Average Treatment Effect (CATE)**: The expected treatment effect conditional on individual characteristics:

$$\tau(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]$$

  Purpose: Captures treatment effect heterogeneity — the fact that the treatment may benefit some units more than others.

- **Unconfoundedness (Ignorability)**: The assumption that treatment assignment is independent of potential outcomes given observed covariates:

$$\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp W_i \mid X_i$$

  Interpretation: Given X_i, treatment is as good as random.

- **Overlap (Positivity)**: Every unit has a non-zero probability of receiving either treatment or control:

$$0 < e(x) = P(W_i = 1 \mid X_i = x) < 1 \quad \forall x$$

  Violation: Units with $e(x) \approx 0$ or $e(x) \approx 1$ have no valid counterfactual comparison group.

- **SUTVA (Stable Unit Treatment Value Assumption)**: Two conditions:
  - **No interference**: Unit i's outcome is unaffected by the treatment of unit j
  - **No hidden versions**: There is only one version of each treatment level

2. Directed Acyclic Graphs (DAGs)

- **DAG (Directed Acyclic Graph)**: A graphical representation of causal assumptions where nodes are variables and directed edges ($\rightarrow$) represent direct causal relationships. Acyclic means no variable causes itself.
- **Confounder**: A variable C that causes both the treatment W and the outcome Y, opening a non-causal (backdoor) path between them. *Effect: Creates spurious association between W and Y in observational data.*
- **Mediator**: A variable M that lies on the directed causal path from W to Y ($W \rightarrow M \rightarrow Y$). *Warning: Conditioning on a mediator blocks the causal pathway you intend to measure — a form of post-treatment bias.*
- **Collider**: A variable C that is a common effect of two other variables ($X \rightarrow C \leftarrow Y$). *Critical Rule: Conditioning on a collider opens a spurious non-causal association between its parents, even if they were originally independent.*
- **Collider Bias (Berkson's Paradox)**: The spurious correlation induced by conditioning on a collider. Example: in a hospital sample, two unrelated diseases appear negatively correlated because hospitalisation is a collider.
- **D-Separation**: A graphical criterion to determine whether two sets of variables X and Y are conditionally independent given a set Z in a DAG.
  - A path is **blocked** if it contains a non-collider conditioned on, or a collider *not* conditioned on
  - $X \perp\!\!\!\perp Y \mid Z$ if all paths between X and Y are blocked by Z
- **Backdoor Criterion**: A set X satisfies the backdoor criterion relative to (W, Y) if:
  1. No variable in X is a descendant of W
  2. X blocks all backdoor paths from W to Y

  Result: Conditioning on X is sufficient for identification of the causal effect.
- **Backdoor Adjustment Formula**:

$$P(Y \mid do(W = w)) = \sum_{x} P(Y \mid W = w, X = x)\, P(X = x)$$

- **Frontdoor Criterion**: A set M satisfies the frontdoor criterion relative to (W, Y) if:
  1. All directed paths from W to Y pass through M
  2. No unblocked backdoor path exists from W to M
  3. All backdoor paths from M to Y are blocked by W

  Advantage: Allows identification of causal effects even with unmeasured confounders, provided a measured mediator satisfies these conditions.

- **Do-Operator** $do(W = w)$: Pearl's notation for an external intervention that sets W to w, removing all incoming arrows to W in the DAG (graph surgery).

$$P(Y \mid do(W = w)) \neq P(Y \mid W = w)$$

- **Markov Equivalence Class (MEC)**: The set of all DAGs that encode the same conditional independence relationships and are therefore indistinguishable from observational data alone. Represented by a CPDAG.

3. Propensity Scores and Weighting

- **Propensity Score** $e(x)$: The conditional probability of receiving treatment given covariates:

$$e(x) = P(W_i = 1 \mid X_i = x)$$

Key Theorem (Rosenbaum & Rubin): If unconfoundedness holds given X, it also holds given $e(X)$ alone — reducing high-dimensional adjustment to one dimension.

- **Inverse Propensity Weighting (IPW)**: Re-weights observed outcomes by the inverse probability of treatment to simulate a randomised experiment:

$$\hat{\tau}_{\text{IPW}} = \frac{1}{n} \sum_i \left[\frac{W_i Y_i}{e(X_i)} - \frac{(1 - W_i) Y_i}{1 - e(X_i)} \right]$$

Limitation: Extreme weights when $e(x) \approx 0$ or $e(x) \approx 1$ cause high variance.

- **Augmented IPW (AIPW) / Doubly Robust Estimator**: Combines outcome regression with IPW:

$$\hat{\tau}_{\text{AIPW}} = \frac{1}{n} \sum_i \left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{W_i (Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)} - \frac{(1 - W_i)(Y_i - \hat{\mu}_0(X_i))}{1 - \hat{e}(X_i)} \right]$$

*Doubly Robust Property: Consistent if **either** the outcome model or the propensity model is correctly specified — not necessarily both.*

- **Propensity Score Matching**: For each treated unit, find the most similar control unit based on $\hat{e}(x)$. Creates a pseudo-experiment. *Limitation: Discards unmatched units, reducing effective sample size.*

- **Propensity Score Stratification**: Divide units into bins (e.g., quintiles) of $\hat{e}(x)$ and estimate treatment effects within each bin, then average. *Advantage: Uses all observations, no units discarded.*

- **Overlap / Common Support**: The region of covariate space where both treated and control units exist. Formally: $e(x) \in (0, 1)$ for all x in the support.

4. Instrumental Variables (IV)

- **Instrument** Z: A variable that satisfies:
  1. **Relevance**: $\text{Cov}(Z, W) \neq 0$ — Z predicts treatment

2. **Exclusion**: Z affects Y only through W — no direct path $Z \to Y$
3. **Exogeneity**: Z is independent of all unmeasured confounders of W and Y

- **Two-Stage Least Squares (2SLS)**: The standard IV estimation procedure:
  - **Stage 1**: Regress W on Z, obtain $\hat{W}$
  - **Stage 2**: Regress Y on $\hat{W}$

$$\hat{\tau}_{IV} = \frac{\text{Cov}(Y, Z)}{\text{Cov}(W, Z)}$$

- **Local Average Treatment Effect (LATE)**: The causal effect identified by IV, applying only to *compliers* — units whose treatment status is changed by the instrument. *Limitation: Not the ATE for the full population; external validity depends on the complier subgroup.*

- **Compliers**: Units for whom $W_i = 1$ if $Z_i = 1$ and $W_i = 0$ if $Z_i = 0$. IV identifies the treatment effect for this subgroup only.

5. Sensitivity Analysis

- **Sensitivity Analysis**: A set of methods to assess how robust causal conclusions are to potential violations of the unconfoundedness assumption. *Core Question: How strong would an unmeasured confounder need to be to overturn the observed findings?*

- **Rosenbaum Bounds**: A framework parameterising the strength of hidden bias by $\Gamma \geq 1$:
  - $\Gamma = 1$: No unmeasured confounding
  - $\Gamma = 2$: An unmeasured confounder doubles the odds of treatment

  Reporting: State the smallest Γ *that renders results statistically insignificant.*

- **Partial Identification (Manski Bounds)**: Under worst-case assumptions about missing counterfactuals, derive bounds on the causal effect:

$$\tau \in [L(\Gamma),\ U(\Gamma)]$$

  Interpretation: If bounds exclude zero even for large Γ*, the conclusion is robust.*

- **E-value (VanderWeele & Ding)**: The minimum strength of association an unmeasured confounder must have with *both* treatment and outcome to fully explain away the observed causal effect estimate.

6. Causal Discovery

- **Causal Discovery**: The task of learning the structure of a causal DAG (or equivalence class) from observational data, without relying on prior domain knowledge alone.

- **Faithfulness Assumption**: Every conditional independence present in the data is also represented as d-separation in the DAG. Without this, the graph cannot be reliably recovered from data.

- **PC Algorithm**: A constraint-based causal discovery algorithm:
  1. Start with a complete undirected graph
  2. Remove edges between conditionally independent variable pairs
  3. Orient v-structures (colliders) where identifiable
  4. Propagate orientations using Meek rules

  Output: A CPDAG representing the full Markov Equivalence Class. Cannot distinguish all edges within an MEC.

- **CPDAG (Completed Partially Directed Acyclic Graph)**: A graph with both directed and undirected edges that represents all DAGs in a Markov Equivalence Class. Directed edges are shared across all members; undirected edges vary.

- **LiNGAM (Linear Non-Gaussian Acyclic Model)**: A functional causal model assuming:

$$X_j = \sum_{k \in \mathrm{pa}(j)} b_{jk} X_k + \varepsilon_j, \qquad \varepsilon_j \sim \text{non-Gaussian, mutually independent}$$

  Key Result: Under non-Gaussian noise, the causal direction is fully identifiable — a unique DAG is recovered, not just an equivalence class.

- **Additive Noise Model (ANM)**: A nonlinear extension of LiNGAM where $Y = f(X) + \varepsilon$ with $\varepsilon \perp\!\!\!\perp X$. Causal direction is identifiable under non-Gaussianity or nonlinearity.

- **V-structure (Collider in Discovery)**: A pattern $X \to Z \leftarrow Y$ where X and Y are not adjacent. V-structures are the only edges that can be oriented from observational data under the faithfulness assumption.

7. Robinson's Decomposition and R-Learning

- **Robinson's Transformation**: A decomposition that isolates the heterogeneous treatment effect through residualisation:

$$Y_i - m(X_i) = \tau(X_i)(W_i - e(X_i)) + \varepsilon_i$$

  where $m(X_i) = \mathbb{E}[Y_i \mid X_i]$ and $e(X_i) = P(W_i = 1 \mid X_i)$. *Significance: Removes baseline outcome variation and propensity confounding, exposing the pure CATE.*

- **R-Loss (Targeting Loss)**: The squared-error objective derived from Robinson's transformation:

$$L(\tau(X_i); X_i, Y_i, W_i) = ((Y_i - m(X_i)) - \tau(X_i)(W_i - e(X_i)))^2$$

  Purpose: Allows arbitrary ML models to directly minimise empirical causal regret.

- **R-Learner**: A meta-learner that (1) estimates nuisance functions $\hat{m}(x)$ and $\hat{e}(x)$ via cross-fitting, then (2) minimises the plug-in R-loss to obtain $\hat{\tau}(x)$.

- **Nuisance Functions**: Intermediate statistical quantities ($m(x)$ and $e(x)$) required for causal identification but not of direct scientific interest. Their estimation quality propagates into CATE estimation.

- **Cross-Fitting**: A sample-splitting strategy where nuisance functions are estimated on a held-out fold and applied to the complementary fold. *Why it matters: Prevents overfitting correlation between residualised outcomes and treatments, ensuring valid inference.*

- **Quasi-Oracle Property**: The asymptotic property where an estimator achieves the same MSE bounds as if the true nuisance parameters were known in advance. *Result: The R-Learner exhibits this property under mild conditions, making it robust to secondary model errors.*

8. Validation and Policy Learning

- **CATE Stacking**: An ensemble method combining multiple CATE estimates $\hat{\tau}_1(x), \ldots, \hat{\tau}_M(x)$ via learned convex weights:

$$\hat{\tau}_\alpha(x) = \sum_{m=1}^{M} \alpha_m \hat{\tau}_m(x), \quad \alpha_m \geq 0, \ \sum_m \alpha_m = 1$$

  Mechanism: Weights are solved via a constrained quadratic program using the out-of-sample R-loss.

- **Calibration Regression**: A diagnostic regressing observed outcomes against predicted CATEs to assess scaling. *Interpretation: Slope $\beta \approx 1$ is ideal. $\beta < 1$ signals overfitted heterogeneity; $\beta > 1$ implies underfitted variation.*

- **Qini Curve**: A visual tool plotting cumulative incremental causal gain against the fraction of the population targeted, sorted by predicted CATE. Used to identify the optimal targeting threshold under budget constraints.

- **Qini Score**: The area under the Qini curve relative to a baseline policy. Quantifies extra value generated by targeted treatment assignment versus random or uniform allocation.

- **Policy Learning**: The task of learning a mapping $\pi : X \to \{0, 1\}$ that maximises aggregate population value:

$$V(\pi) = \mathbb{E}[Y_i(\pi(X_i))]$$

  Distinction: CATE estimation is descriptive (how large is the effect?); policy learning is prescriptive (should we treat?).

- **Doubly Robust Policy Loss (AIPW)**: A robust objective for policy evaluation:

$$\hat{V}_{\text{AIPW}}(\pi) = \frac{1}{n} \sum_i (2\pi(X_i) - 1)\hat{\Gamma}_i$$

  where $\hat{\Gamma}_i$ is the doubly robust pseudo-outcome score.

- **Weighted Classification Reduction**: The reformulation of policy optimisation as a cost-sensitive binary classification task. Units receive labels $\text{sign}(\hat{\Gamma}_i)$ and weights $|\hat{\Gamma}_i|$.

- **Treatment Cost Threshold (C)**: The marginal cost of administering an intervention. Optimal policy: treat unit i if and only if $\tau(x) > C$.

Common Shortnames

Shortname	Meaning
ATE	Average Treatment Effect
CATE	Conditional Average Treatment Effect
LATE	Local Average Treatment Effect
DAG	Directed Acyclic Graph
CPDAG	Completed Partially Directed Acyclic Graph
MEC	Markov Equivalence Class
IV	Instrumental Variable
2SLS	Two-Stage Least Squares
IPW	Inverse Propensity Weighting
AIPW	Augmented Inverse Propensity Weighting
SUTVA	Stable Unit Treatment Value Assumption
ML	Machine Learning
R-Loss	Residualised Loss Function (Robinson-based targeting loss)
R-Learner	Residualised Meta-Learner
MSE	Mean Squared Error
ANM	Additive Noise Model
LiNGAM	Linear Non-Gaussian Acyclic Model
PC Algorithm	Peter-Clark Causal Discovery Algorithm
$e(x)$	Propensity score: $P(W = 1 \mid X = x)$
$m(x)$	Conditional outcome mean: $\mathbb{E}[Y \mid X = x]$
$\hat{\Gamma}_i$	Doubly robust pseudo-outcome score for unit i
$\tau(x)$	CATE function
$\pi(x)$	Policy function: $X \to \{0, 1\}$
$V(\pi)$	Expected value of policy π
$do(W = w)$	Pearl's intervention operator